Discovery Lab

Exam 6 Revision

Lower Secondary Stage 2 · Science · Lessons 29 to 33

Forces and springs, friction and speed, pressure, variation and natural selection, and adaptations.



Lesson 29: Force and Mass

Lesson 30: Friction and Sport

Lesson 31: Pressure

Lesson 32: Variation in Animals

Lesson 33: Adaptations and Environmental Effects

Name: _____ Class: _____ Date: _____

LESSON 29

29 Force and Mass

Pushes, pulls, weight and springs.

Key ideas

A **force** is a push or pull, measured in **newtons (N)** with a newton meter. It can start or stop movement, change direction or change shape.

Mass (kg) is the amount of matter and never changes. **Weight** (N) is the force of gravity: **W = m × g** (g = 10 N/kg on Earth, 1.6 on the Moon).

Hooke's law: **F = k × e**. Extension is proportional to force until the **elastic limit**; the graph is a straight line through the origin.

Types of force: gravity, friction, air resistance, magnetic force, tension.

Force (N)	Extension (cm)
0	0
1	2
2	4
3	6
4	8
5	10

DID YOU KNOW?

On the Moon a person who weighs 600 N on Earth weighs only 96 N, but their mass is the same.

Check yourself

- [ ] What is a force and what is it measured in?
- [ ] What is the weight of a 5 kg object on Earth?
- [ ] What does k mean in F = k × e?
- [ ] What happens beyond the elastic limit?
- [ ] Why does weight change on the Moon but mass does not?

REVISION TASK

Draw a Hooke's law graph with force on the y-axis and extension on the x-axis, and mark the elastic limit.

LESSON 30

30 Friction and Sport

Resisting forces, balanced forces and speed.

Key ideas

Friction opposes motion between surfaces and makes heat. Air and water give **air resistance** and **water resistance**.

Low friction wanted: swimming, skiing, ice skating. High friction wanted: running, rock climbing, football.

Balanced forces: no change in motion. **Unbalanced** forces: speed or direction changes. Forward force > friction speeds up; friction > forward force slows down.

Speed = distance ÷ time (m/s). Speed-time graph: flat = constant speed, rising = acceleration, falling = deceleration, flat at 0 = stopped.

Situation	Balanced?	Speed change
Car at a steady 50 km/h	Balanced	Constant speed
Ball slowing on grass	Unbalanced	Slows (friction)
Rocket launching	Unbalanced	Speeds up
Cyclist coasting downhill	Unbalanced	Speeds up

DID YOU KNOW?

In curling, athletes sweep the ice to reduce friction and control the stone.

Check yourself

- [ ] What is friction and what does it produce?
- [ ] Name two sports that want low friction.
- [ ] What are unbalanced forces?
- [ ] A runner covers 100 m in 20 s. What is her speed?
- [ ] What does a flat line above zero on a speed-time graph show?

REVISION TASK

Draw a speed-time graph for a car that speeds up, travels at a steady speed and then stops.

LESSON 31

31 Pressure

The same force can press hard or gently.

Key ideas

Pressure = force ÷ area, in **pascals (Pa)**. More force or smaller area gives more pressure.

Snowshoes spread your weight over a large area, so you do not sink. **High heels** concentrate it on a tiny area.

A person of **600 N** on **0.1 m²** gives 600 ÷ 0.1 = **6000 Pa**.

Tyres: under-inflated = too much contact, more friction and wear; over-inflated = too little contact, less grip; correct pressure = safe grip and even wear.

Term	Meaning
Pressure	Force acting over an area
Force	A push or pull (N)
Area	The surface the force acts on (m^2)
Pascal (Pa)	Unit of pressure
Newton (N)	Unit of force

DID YOU KNOW?

An elephant's footprints can press less than a woman in high heels because its weight is spread over huge feet.

Check yourself

- [ ] What is the formula for pressure?
- [ ] What increases pressure?
- [ ] Why do snowshoes stop you sinking?
- [ ] 90 N acts on 0.3 m^2. What is the pressure?
- [ ] What happens when a tyre is over-inflated?

REVISION TASK

Draw a high-heel foot and a snowshoe and label the area and the pressure of each.

LESSON 32

32 Variation in Animals

No two individuals are exactly the same.

Key ideas

Inherited variation comes from DNA (eye colour, blood group). **Environmental** variation comes from surroundings (scars, accent). **Both**: height and mass.

Continuous variation is a range of values (height, mass, foot length). **Discontinuous** variation fits set categories (blood group, tongue rolling).

Variation helps a species survive change. **Natural selection**: individuals with helpful traits survive and reproduce more.

Artificial selection (selective breeding): people choose the parents, for example the fastest horses.

Trait	Type
Eye colour	Inherited
A scar	Environmental
Height	Both
Blood group	Discontinuous
Arm span	Continuous

DID YOU KNOW?

Atlantic cod fell by about 90% in some areas, so scientists study genetic variation to rebuild stronger fish stocks.

Check yourself

- [ ] What is variation?
- [ ] Give one inherited and one environmental human characteristic.
- [ ] What is the difference between continuous and discontinuous variation?
- [ ] What is natural selection?
- [ ] Why is variation important?

REVISION TASK

Make a table with two examples each of inherited, environmental, continuous and discontinuous variation.

LESSON 33

33 Adaptations and Environmental Effects

How living things fit their habitat, and what happens when it changes.

Key ideas

An **adaptation** helps an organism survive. A swordfish has a **streamlined body**, **long bill**, **large eyes**, **powerful tail** and **counter-shading** (dark top, light belly).

A **habitat** is where an organism finds food, water and shelter.

Natural changes: drought, flood, volcano. Human-made: pollution, deforestation, urban development, farming, climate change.

Cause and effect: building cities → frogs lose clean water; oil spill → fish and birds harmed; climate warms → polar bears lose ice; deforestation → birds lose nests.

Adaptation	How it helps
Streamlined body	Reduces water resistance
Long bill	Catches fast prey
Large eyes	Sees in deep, dark water
Counter-shading	Camouflage from above and below
Powerful tail	High-speed swimming

DID YOU KNOW?

Swordfish can warm their brain and eyes to see better when hunting.

Check yourself

- [ ] What is an adaptation?
- [ ] Name two adaptations of a swordfish.
- [ ] What is counter-shading?
- [ ] How does deforestation harm animals?
- [ ] How does natural selection help a species survive?

REVISION TASK

Draw a swordfish and label four adaptations with how each one helps it survive.

FINAL PRACTICE

Final practice

Tick each box when you can answer without looking.

Final practice

- [ ] Explain the difference between mass and weight, including units and relationship.
- [ ] Describe useful and unwanted friction in sport and how athletes manage them.
- [ ] Explain why high heels cause more pressure than flat shoes.
- [ ] Explain how natural selection helps species survive in changing environments.
- [ ] Calculate the speed of a cyclist who travels 150 m in 30 s.

CHALLENGE

Link two topics: explain how friction and pressure both affect a car's tyres.

Revision checklist

Revision task	Done?
Reviewed the Canvas slides for Lessons 29–33	☐
Answered every "Check yourself" question	☐
Practised the matching and table activities	☐
Used the 3D labs: springs, friction, pressure and natural selection	☐
Practised short answers and the practice exam	☐
Reviewed the keywords: Hooke's law, friction, pressure, natural selection	☐

Try it live · Practise with 3D labs, games and a practice exam at **morehumaned.com/discovery-lab**

ANSWER KEY

Answer key

Check your own answers against these. If one is hard, go back to the lesson.

Lesson 29: Force and Mass

1. A push or pull, in newtons.
2. 5 × 10 = 50 N.
3. The spring constant (stiffness) in N/m.
4. The spring does not return to its original shape.
5. Gravity is weaker on the Moon, but the amount of matter is the same.

Lesson 30: Friction and Sport

1. A force that opposes motion; it produces heat.
2. For example swimming and ice skating.
3. Forces that change an object's speed or direction.
4. 100 ÷ 20 = 5 m/s.
5. Constant speed.

Lesson 31: Pressure

1. Pressure = force ÷ area.
2. More force or a smaller area.
3. They spread the weight over a larger area, so the pressure is lower.
4. 90 ÷ 0.3 = 300 Pa.
5. Less of the tyre touches the road, so there is less grip.

Lesson 32: Variation in Animals

1. The differences between individuals in a species.
2. Eye colour is inherited; a scar is environmental.
3. Continuous is a range of values; discontinuous fits set categories.
4. Individuals with helpful traits survive and reproduce more.
5. Some individuals may survive if the environment changes.

Lesson 33: Adaptations and Environmental Effects

1. A feature that helps an organism survive in its habitat.
2. For example a streamlined body and a long bill.
3. A dark top and a light belly for camouflage.
4. Animals lose their nests, food and shelter.
5. Individuals with helpful traits survive and pass them on.

Final practice

1. Mass is the amount of matter (kg); weight is the force of gravity (N); W = m × g.
2. Friction grips (trainers) or slows (water); athletes add grip or streamline and lubricate.
3. The same weight acts on a much smaller area, so the pressure is higher.
4. Helpful traits help individuals survive, reproduce and pass them on over generations.
5. 150 ÷ 30 = 5 m/s.

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You are going to do great! Use this booklet to plan your study time, then bring it all together.

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